

# SAFETY DATA SHEET

Creation Date 04-Oct-2012

Revision Date 10-Dec-2021

Revision Number 2

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: **Salmonella H 1,7 Agglutinating Sera**  
Cat No. : **R30163601**

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use In vitro diagnostic.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

Company Oxoid Ltd  
Wade Road  
Basingstoke, Hants, UK  
RG24 8PW  
Tel: +44 (0) 1256 841144

**EU entity/business name**

Oxoid Deutschland GmbH  
Postfach 10 07 53  
D-46483  
Wesel  
GERMANY  
Tel: + 49 (0) 281 1520  
Fax: 49 (0) 281 1521

E-mail address mbd-sds@thermofisher.com

### 1.4. Emergency telephone number

Chemtrec EU: 001-703-527-3887  
Chemtrec US: (800) 424-9300

For customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - Regulation (EC) No 1272/2008

#### Physical hazards

Based on available data, the classification criteria are not met

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## Health hazards

Based on available data, the classification criteria are not met

## Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements

None required

Signal Word

None

## 2.3. Other hazards

No information available

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component        | CAS No     | EC No     | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                     |
|------------------|------------|-----------|----------|---------------------------------------------------------------------------------------|
| Sodium hydroxide | 1310-73-2  | 215-185-5 | <0.5     | Skin Corr. 1A (H314)<br>Eye Dam. 1 (H318)                                             |
| Sodium azide     | 26628-22-8 | 247-852-1 | 0.1      | Acute Tox. 2 (H300)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410)<br>(EUH032) |

| Component        | Specific concentration limits (SCL's)                                                                            | M-Factor | Component notes |
|------------------|------------------------------------------------------------------------------------------------------------------|----------|-----------------|
| Sodium hydroxide | Skin Corr. 1A :: C>=5%<br>Skin Corr. 1B :: 2%<=C<5%<br>Eye Irrit. 2 :: 0.5%<=C<2%<br>Skin Irrit. 2 :: 0.5%<=C<2% | -        | -               |
| Sodium azide     | -                                                                                                                | 1        | -               |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

Eye Contact

Rinse thoroughly with plenty of water, also under the eyelids. Get medical attention.

Skin Contact

Get medical attention. Wash with plenty of soap and water.

Ingestion

Do NOT induce vomiting. Get medical attention.

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|                                    |                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation                         | Remove to fresh air. Get medical attention.                                                                                                      |
| Self-Protection of the First Aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

## **4.2. Most important symptoms and effects, both acute and delayed**

No information available.

## **4.3. Indication of any immediate medical attention and special treatment needed**

Notes to Physician Treat symptomatically.

## **SECTION 5: FIREFIGHTING MEASURES**

### **5.1. Extinguishing media**

#### **Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### **5.2. Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors.

#### **Hazardous Combustion Products**

None under normal use conditions.

### **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **6.1. Personal precautions, protective equipment and emergency procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing.

### **6.2. Environmental precautions**

Prevent further leakage or spillage if safe to do so. Prevent product from entering drains.

### **6.3. Methods and material for containment and cleaning up**

Soak up with inert absorbent material. Clean contaminated surface thoroughly.

### **6.4. Reference to other sections**

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

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## 7.1. Precautions for safe handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Use only in well-ventilated areas.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 7.2. Conditions for safe storage, including any incompatibilities

Keep container tightly closed. Keep at temperatures between 2° and 8 °C.

Technical Rules for Hazardous Substances (TRGS) 510  
Storage Class (LGK) (Germany)

Storage Class/LGK 12

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund). **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC

| Component        | European Union                                                  | The United Kingdom                                              | France                                                                                                                           | Belgium                                                         | Spain                                                                                                       |
|------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Sodium hydroxide |                                                                 | 2 mg/m <sup>3</sup> STEL                                        | TWA / VME: 2 mg/m <sup>3</sup> (8 heures).                                                                                       | 2 mg/m <sup>3</sup> VLE                                         | STEL / VLA-EC: 2 mg/m <sup>3</sup> (15 minutos).                                                            |
| Sodium azide     | Skin<br>TWA 0.1 mg/m <sup>3</sup><br>STEL 0.3 mg/m <sup>3</sup> | Skin<br>TWA 0.1 mg/m <sup>3</sup><br>STEL 0.3 mg/m <sup>3</sup> | TWA / VME: 0.1 mg/m <sup>3</sup> (8 heures). restrictive limit<br>STEL / VLCT: 0.3 mg/m <sup>3</sup> . restrictive limit<br>Peau | Skin<br>TWA 0.1 mg/m <sup>3</sup><br>STEL 0.3 mg/m <sup>3</sup> | STEL / VLA-EC: 0.3 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 0.1 mg/m <sup>3</sup> (8 horas)<br>Piel |

| Component        | Italy                                                                                                                        | Germany                                      | Portugal                                                                                                                                     | The Netherlands                                                                     | Finland                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Sodium hydroxide |                                                                                                                              | 2 mg/m <sup>3</sup> TWA (inhalable fraction) | Ceiling: 2 mg/m <sup>3</sup>                                                                                                                 |                                                                                     | Ceiling: 2 mg/m <sup>3</sup>                                                               |
| Sodium azide     | TWA: 0.1 mg/m <sup>3</sup> 8 ore.<br>Media Ponderata nel Tempo<br>STEL: 0.3 mg/m <sup>3</sup> 15 minuti. Breve termine Pelle | MAK 0.2 mg/m <sup>3</sup> (inhalable)        | STEL: 0.3 mg/m <sup>3</sup> 15 minutos<br>Ceiling: 0.29 mg/m <sup>3</sup><br>Ceiling: 0.11 ppm<br>TWA: 0.1 mg/m <sup>3</sup> 8 horas<br>Pele | huid<br>STEL: 0.3 mg/m <sup>3</sup> 15 minuten<br>TWA: 0.1 mg/m <sup>3</sup> 8 uren | TWA: 0.1 mg/m <sup>3</sup> 8 tunteina<br>STEL: 0.3 mg/m <sup>3</sup> 15 minuutteina<br>Iho |

| Component        | Austria                                                                                        | Denmark                                   | Switzerland                                                                    | Poland                                                                            | Norway                                                                                                   |
|------------------|------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Sodium hydroxide | MAK-KZGW: 4 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 2 mg/m <sup>3</sup> 8 Stunden             | Ceiling: 2 mg/m <sup>3</sup>              | STEL: 2 mg/m <sup>3</sup> 15 Minuten<br>TWA: 2 mg/m <sup>3</sup> 8 Stunden     | STEL: 1 mg/m <sup>3</sup> 15 minutach<br>TWA: 0.5 mg/m <sup>3</sup> 8 godzinach   | Ceiling: 2 mg/m <sup>3</sup>                                                                             |
| Sodium azide     | Haut<br>MAK-KZGW: 0.3 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 0.1 mg/m <sup>3</sup> 8 Stunden | TWA: 0.1 mg/m <sup>3</sup> 8 timer<br>Hud | STEL: 0.4 mg/m <sup>3</sup> 15 Minuten<br>TWA: 0.2 mg/m <sup>3</sup> 8 Stunden | STEL: 0.3 mg/m <sup>3</sup> 15 minutach<br>TWA: 0.1 mg/m <sup>3</sup> 8 godzinach | TWA: 0.1 mg/m <sup>3</sup> 8 timer<br>STEL: 0.3 mg/m <sup>3</sup> 15 minutter. value from the regulation |

| Component | Bulgaria | Croatia | Ireland | Cyprus | Czech Republic |
|-----------|----------|---------|---------|--------|----------------|
|-----------|----------|---------|---------|--------|----------------|

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|                  |                                                                             |                                                                                                   |                                                                                |                                                                                                      |                                                                                                                |
|------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Sodium hydroxide | TWA: 2.0 mg/m <sup>3</sup>                                                  | STEL-KGVI: 2 mg/m <sup>3</sup> 15 minutama.                                                       | STEL: 2 mg/m <sup>3</sup> 15 min                                               |                                                                                                      | TWA: 1 mg/m <sup>3</sup> 8 hodinách.<br>Ceiling: 2 mg/m <sup>3</sup>                                           |
| Sodium azide     | TWA: 0.1 mg/m <sup>3</sup><br>STEL : 0.3 mg/m <sup>3</sup><br>Skin notation | kože<br>TWA-GVI: 0.1 mg/m <sup>3</sup> 8 satima.<br>STEL-KGVI: 0.3 mg/m <sup>3</sup> 15 minutama. | TWA: 0.1 mg/m <sup>3</sup> 8 hr.<br>STEL: 0.3 mg/m <sup>3</sup> 15 min<br>Skin | Skin-potential for cutaneous absorption<br>STEL: 0.3 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.1 mg/m <sup>3</sup> 8 hodinách.<br>Potential for cutaneous absorption<br>Ceiling: 0.3 mg/m <sup>3</sup> |

| Component        | Estonia                                                                                     | Gibraltar                                                                              | Greece                                                                                     | Hungary                                                                                 | Iceland                                                                                     |
|------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Sodium hydroxide | TWA: 1 mg/m <sup>3</sup> 8 tundides.<br>STEL: 2 mg/m <sup>3</sup> 15 minutites.             |                                                                                        | STEL: 2 mg/m <sup>3</sup><br>TWA: 2 mg/m <sup>3</sup>                                      | STEL: 2 mg/m <sup>3</sup> 15 percekben. CK<br>TWA: 1 mg/m <sup>3</sup> 8 órában. AK     | STEL: 2 mg/m <sup>3</sup>                                                                   |
| Sodium azide     | Nahk<br>TWA: 0.1 mg/m <sup>3</sup> 8 tundides.<br>STEL: 0.3 mg/m <sup>3</sup> 15 minutites. | Skin notation<br>TWA: 0.1 mg/m <sup>3</sup> 8 hr<br>STEL: 0.3 mg/m <sup>3</sup> 15 min | STEL: 0.1 ppm<br>STEL: 0.3 mg/m <sup>3</sup><br>TWA: 0.1 ppm<br>TWA: 0.3 mg/m <sup>3</sup> | STEL: 0.3 mg/m <sup>3</sup> 15 percekben. CK<br>TWA: 0.1 mg/m <sup>3</sup> 8 órában. AK | STEL: 0.3 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> 8 klukkustundum.<br>Skin notation |

| Component        | Latvia                                                                                               | Lithuania                                                             | Luxembourg                                                                                                                           | Malta                                                                                                                     | Romania                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Sodium hydroxide | TWA: 0.5 mg/m <sup>3</sup>                                                                           | Ceiling: 2 mg/m <sup>3</sup>                                          |                                                                                                                                      |                                                                                                                           |                                                                                            |
| Sodium azide     | skin - potential for cutaneous exposure<br>STEL: 0.3 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.1 mg/m <sup>3</sup> IPRD<br>Oda<br>STEL: 0.3 mg/m <sup>3</sup> | Possibility of significant uptake through the skin<br>TWA: 0.1 mg/m <sup>3</sup> 8 Stunden<br>STEL: 0.3 mg/m <sup>3</sup> 15 Minuten | possibility of significant uptake through the skin<br>TWA: 0.1 mg/m <sup>3</sup><br>STEL: 0.3 mg/m <sup>3</sup> 15 minuti | Skin notation<br>TWA: 0.1 mg/m <sup>3</sup> 8 ore<br>STEL: 0.3 mg/m <sup>3</sup> 15 minute |

| Component        | Russia | Slovak Republic                                                                                    | Slovenia                                                                            | Sweden                                                                                     | Turkey                                                                             |
|------------------|--------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Sodium hydroxide |        | TWA: 2 mg/m <sup>3</sup>                                                                           |                                                                                     | Binding STEL: 2 mg/m <sup>3</sup> 15 minuter KGV<br>TLV: 1 mg/m <sup>3</sup> 8 timmar. NGV |                                                                                    |
| Sodium azide     |        | Ceiling: 0.3 mg/m <sup>3</sup><br>Potential for cutaneous absorption<br>TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.1 mg/m <sup>3</sup> 8 urah<br>Koža<br>STEL: 0.3 mg/m <sup>3</sup> 15 minutah | Binding STEL: 0.3 mg/m <sup>3</sup> 15 minuter<br>TLV: 0.1 mg/m <sup>3</sup> 8 timmar. NGV | Deri<br>TWA: 0.1 mg/m <sup>3</sup> 8 saat<br>STEL: 0.3 mg/m <sup>3</sup> 15 dakika |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                          | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|------------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Sodium azide<br>26628-22-8 ( 0.1 ) |                              |                                 |                                | DNEL = 46.7µg/kg bw/day           |

| Component        | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Sodium hydroxide |                                  |                                     | DNEL = 1mg/m <sup>3</sup>          |                                       |

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|                                    |  |  |  |                               |
|------------------------------------|--|--|--|-------------------------------|
| 1310-73-2 ( <0.5 )                 |  |  |  |                               |
| Sodium azide<br>26628-22-8 ( 0.1 ) |  |  |  | DNEL = 0.164mg/m <sup>3</sup> |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                          | Fresh water     | Fresh water sediment         | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture) |
|------------------------------------|-----------------|------------------------------|--------------------|------------------------------------|--------------------|
| Sodium azide<br>26628-22-8 ( 0.1 ) | PNEC = 0.35µg/L | PNEC = 16.7µg/kg sediment dw | PNEC = 3.5µg/L     | PNEC = 30µg/L                      |                    |

| Component                          | Marine water  | Marine water sediment        | Marine water Intermittent | Food chain | Air |
|------------------------------------|---------------|------------------------------|---------------------------|------------|-----|
| Sodium azide<br>26628-22-8 ( 0.1 ) | PNEC = 15ng/L | PNEC = 0.72µg/kg sediment dw | PNEC = 150ng/L            |            |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material    | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Disposable gloves | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

In case of insufficient ventilation, wear suitable respiratory equipment

#### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

Prevent product from entering drains.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

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|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Physical State                          | Liquid                   |                                   |
| Appearance                              | Amber                    |                                   |
| Odor                                    | No information available |                                   |
| Odor Threshold                          | No data available        |                                   |
| Melting Point/Range                     | Not applicable           |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | Not applicable           |                                   |
| Flammability (liquid)                   | No data available        |                                   |
| Flammability (solid,gas)                | No information available |                                   |
| Explosion Limits                        | No data available        |                                   |
| Flash Point                             | Not applicable           | Method - No information available |
| Autoignition Temperature                | Not applicable           |                                   |
| Decomposition Temperature               | No data available        |                                   |
| pH                                      | 6.6 - 6.8                |                                   |
| Viscosity                               | No data available        |                                   |
| Water Solubility                        | No information available |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Vapor Pressure                          | No data available        |                                   |
| Density / Specific Gravity              | No data available        |                                   |
| Bulk Density                            | No data available        |                                   |
| Vapor Density                           | No data available        | (Air = 1.0)                       |
| Particle characteristics                | Not applicable (liquid)  |                                   |

## 9.2. Other information

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under recommended storage conditions.

### 10.3. Possibility of hazardous reactions

#### Hazardous Polymerization

Hazardous polymerization does not occur.

#### Hazardous Reactions

None under normal processing.

### 10.4. Conditions to avoid

Extremes of temperature and direct sunlight.

### 10.5. Incompatible materials

Strong oxidizing agents. Acids.

### 10.6. Hazardous decomposition products

None under normal use conditions.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

Product does not present an acute toxicity hazard based on known information

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**(a) acute toxicity;**

Oral No data available  
Dermal No data available  
Inhalation No data available

| Component        | LD50 Oral                | LD50 Dermal                  | LC50 Inhalation                       |
|------------------|--------------------------|------------------------------|---------------------------------------|
| Sodium hydroxide | LD50 = 325 mg/kg ( Rat ) | LD50 = 1350 mg/kg ( Rabbit ) | -                                     |
| Sodium azide     | LD50 = 27 mg/kg ( Rat )  | -                            | LC50 0.054 - 0.52 mg/L ( Rat )<br>4 h |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

**(d) respiratory or skin sensitization;**

Respiratory No data available  
Skin No data available  
None known

**(e) germ cell mutagenicity;** No data available

None known

**(f) carcinogenicity;** No data available

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** No data available  
Reproductive Effects None known.  
Developmental Effects None known.

**(h) STOT-single exposure;** No data available

**(i) STOT-repeated exposure;** No data available

Target Organs No information available.

**(j) aspiration hazard;** No data available

**Symptoms / effects, both acute and delayed** No information available.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity



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## Ecotoxicity effects

| Component        | Freshwater Fish                                                                                                                                         | Water Flea | Freshwater Algae |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|
| Sodium hydroxide | LC50: = 45.4 mg/L, 96h static (Oncorhynchus mykiss)                                                                                                     |            | -                |
| Sodium azide     | LC50: = 0.7 mg/L, 96h (Lepomis macrochirus)<br>LC50: = 0.8 mg/L, 96h (Oncorhynchus mykiss)<br>LC50: = 5.46 mg/L, 96h flow-through (Pimephales promelas) |            |                  |

| Component        | Microtox | M-Factor |
|------------------|----------|----------|
| Sodium hydroxide | -        |          |
| Sodium azide     |          | 1        |

**12.2. Persistence and degradability** No information available

**12.3. Bioaccumulative potential** No information available

**12.4. Mobility in soil** No information available

**12.5. Results of PBT and vPvB assessment** No data available for assessment.

## 12.6. Endocrine disrupting properties

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products** Dispose of in accordance with local regulations.

**Contaminated Packaging** Empty containers should be taken to an approved waste handling site for recycling or disposal.

**European Waste Catalogue (EWC)** According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used.

**Switzerland - Waste Ordinance** Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600

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<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO** Not regulated

**14.1. UN number**

**14.2. UN proper shipping name**

**14.3. Transport hazard class(es)**

**14.4. Packing group**

**ADR** Not regulated

**14.1. UN number**

**14.2. UN proper shipping name**

**14.3. Transport hazard class(es)**

**14.4. Packing group**

**IATA** Not regulated

**14.1. UN number**

**14.2. UN proper shipping name**

**14.3. Transport hazard class(es)**

**14.4. Packing group**

**14.5. Environmental hazards** No hazards identified

**14.6. Special precautions for user** No special precautions required

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

**15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture**

### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component        | CAS No     | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|------------------|------------|-----------|--------|-----|-------|------|----------|------|------|
| Sodium hydroxide | 1310-73-2  | 215-185-5 | -      | -   | X     | X    | KE-31487 | X    | X    |
| Sodium azide     | 26628-22-8 | 247-852-1 | -      | -   | X     | X    | KE-31357 | X    | X    |

| Component        | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|------------------|------------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Sodium hydroxide | 1310-73-2  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Sodium azide     | 26628-22-8 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High |
|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|

# SAFETY DATA SHEET

Salmonella H 1,7 Agglutinating Sera

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|                  |   |                                                                    |                |
|------------------|---|--------------------------------------------------------------------|----------------|
|                  |   |                                                                    | Concern (SVHC) |
| Sodium hydroxide | - | Use restricted. See item 75.<br>(see link for restriction details) | -              |

<https://echa.europa.eu/substances-restricted-under-reach>

| Component        | CAS No     | Seveso III Directive (2012/18/EC) -<br>Qualifying Quantities for Major<br>Accident Notification | Seveso III Directive (2012/18/EC) -<br>Qualifying Quantities for Safety<br>Report Requirements |
|------------------|------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Sodium hydroxide | 1310-73-2  | Not applicable                                                                                  | Not applicable                                                                                 |
| Sodium azide     | 26628-22-8 | Not applicable                                                                                  | Not applicable                                                                                 |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work. Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work.

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = 1 (self classification)

| Component        | Germany - Water Classification (VwVwS) | Germany - TA-Luft Class |
|------------------|----------------------------------------|-------------------------|
| Sodium hydroxide | WGK1                                   |                         |
| Sodium azide     | WGK2                                   |                         |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                              | Switzerland - Ordinance on the<br>Reduction of Risk from<br>handling of hazardous<br>substances preparation (SR<br>814.81) | Switzerland - Ordinance on<br>Incentive Taxes on Volatile<br>Organic Compounds (OVOC) | Switzerland - Ordinance of the<br>Rotterdam Convention on the<br>Prior Informed Consent<br>Procedure |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Sodium hydroxide<br>1310-73-2 ( <0.5 ) | Prohibited and Restricted<br>Substances                                                                                    |                                                                                       |                                                                                                      |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H300 - Fatal if swallowed

H400 - Very toxic to aquatic life

H410 - Very toxic to aquatic life with long lasting effects

EUH032 - Contact with acids liberates very toxic gas

H314 - Causes severe skin burns and eye damage

H318 - Causes serious eye damage

# SAFETY DATA SHEET

Salmonella H 1,7 Agglutinating Sera

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## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Prepared By**

Regulatory Affairs

**Creation Date**

04-Oct-2012

**Revision Date**

10-Dec-2021

**Revision Summary**

Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**